

CS 105: Department Introductory Course on Discrete Structures

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Sep 09, 2025

Lecture 11 – Basic structures: chains and anti-chains

Logistics

Optional help session tomorrow: at 3.30pm

- ▶ Problemsheet 6
- ▶ Last year midsem?
- ▶ Another basics session this Saturday

Harder questions on piazza?

Chains and Anti-chains

A relation $\preceq$ on set S is called a **partial-order** if it is reflexive, transitive and anti-symmetric. The pair $(S, \preceq)$ is then called a poset!

Definition

Let $(S, \preceq)$ be a poset. A subset $B \subseteq S$ is called

- ▶ a **chain** if every pair of elements in B is related by $\preceq$.
- ▶ That is, $\forall a, b \in B$, we have $a \preceq b$ or $b \preceq a$ (or both).
- ▶ Thus, $\preceq$ is a total order on B .

Definition

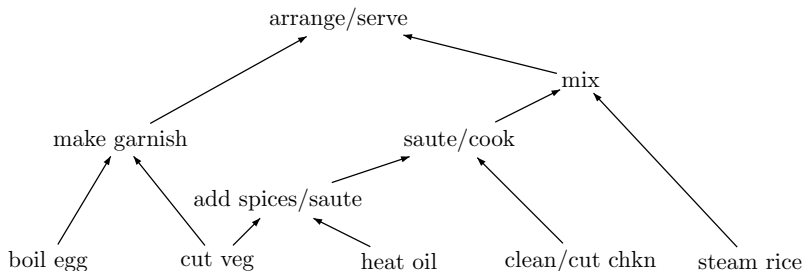
Let $(S, \preceq)$ be a poset. A subset $A \subseteq S$ is called

- ▶ an **anti-chain** if no two distinct elements of A are related by $\preceq$.
- ▶ That is, $\forall a, b \in A, a \neq b$, we have neither $a \preceq b$ nor $b \preceq a$.

Examples and applications

A task scheduling example

Let us represent a recipe for making Chicken Biryani as a poset!



- ▶ **Dependencies** give rise to a poset
- ▶ But for cooking you need a total order, that is consistent with the po.
- ▶ This is called a **linearization** or a **topological sorting**.

Topological sorting

Definition

A **topological sort** or a **linearization** of a poset $(S, \preceq)$ is a poset $(S, \preceq_t)$ with a total order $\preceq_t$ which is **consistent** with the partial order $\preceq$, (i.e., $x \preceq y$ implies $x \preceq_t y$).

Theorem

Every non-empty finite poset has a topological sort.

Minimal (and maximal?) elements

We call x as minimal in S if there exists no $y \in S, y \prec x$ (where, $y \prec x$ is shorthand for $y \preceq x, y \neq x$)

Lemma

Every finite non-empty poset has at least one minimal element.

Proof: Suppose that the poset has k elements.

- ▶ Choose x_1 from the poset - either it is minimal, or there is some $x_2 \prec x_1$ (i.e., $x_2 \neq x_1$ s.t. $x_2 \preceq x_1$).
- ▶ If x_2 is minimal, we are done; otherwise there is some $x_3 \prec x_2$.
- ▶ Repeating this step $k + 1$ times, we get

$$x_{k+1} \prec x_k \prec \cdots \mathbf{x_j} \prec \cdots \prec x_{i+1} \prec \mathbf{x_i} \prec \cdots \prec x_2 \prec x_1$$

- ▶ Since size of poset is k , we must have $x_j = x_i, j < i$.
We get $x_j \preceq x_{i+1}$ (by transitivity) and $x_{i+1} \prec x_i = x_j$ (violates anti-symmetry) Contradiction!

Proof by induction?(H.W)

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What about infinite posets?

Topological sort

Theorem

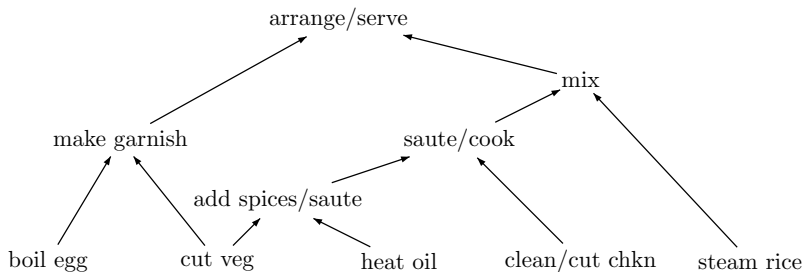
Every finite non-empty poset $(S, \preceq)$ has a topological sort.

Proof:

- ▶ We use the Lemma
 - ▶ Every finite non-empty poset has at least one minimal element.
- ▶ Let S have n elements. Give an algo for constructing the chain! Here, we give an inductive construction:
 - ▶ Start with x_1 , a minimal element of S . This is a chain consistent with $\preceq$.
 - ▶ Suppose we constructed till some $1 \leq k < n$, i.e, we have a chain $x_1 \preceq_t x_2 \dots \preceq_t x_k$, where $\preceq_t$ is consistent with $\preceq$.
 - ▶ Then consider $S \setminus \{x_1, \dots, x_k\}$. Call its minimal element x_{k+1} .
 - ▶ Then $x_1 \preceq_t \dots \preceq_t x_k \preceq_t x_{k+1}$ is chain consistent $\preceq$. If not, $\exists 1 \leq i \leq k$ such that $x_{k+1} \preceq x_i$ but $x_i \not\preceq_t x_{k+1}$. But then at i^{th} step x_i violated minimality!
 - ▶ Thus, after n steps we get $x_1 \preceq_t \dots \preceq_t x_n$ is a chain that is consistent with $\preceq$. □

Topological sort of Biryani Recipe

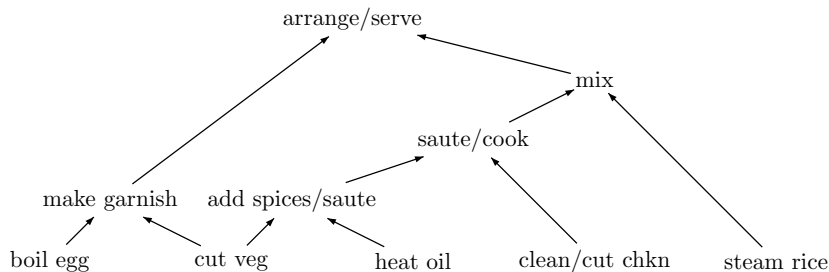
Run the construction on this poset:



Parallel Task Scheduling and chains

Coming back to our example,

- ▶ What if there are many cooks, i.e., parallel processors?
- ▶ How do we schedule the tasks to minimize time used?



- ▶ Assume that every task takes 1 time unit.
- ▶ Clearly, we still need at least 5 time units.
- ▶ That is, the size of the largest/longest chain (size of chain = no. of elements in it).

Parallel task scheduling

For any (non-empty, finite) poset, there is a legal parallel schedule that runs in t steps, where t is the size of the longest chain.

We will in fact prove:

Theorem

For a non-empty, finite poset $(S, \preceq)$ with size of longest chain $= t$, we can partition S into t subsets $S_1, \dots, S_t$ such that $\forall i \in \{1, \dots, t\}, \forall a \in S_i$, if $b \preceq a, b \neq a$ then $b \in S_1 \cup \dots \cup S_{i-1}$.

Assuming this theorem,

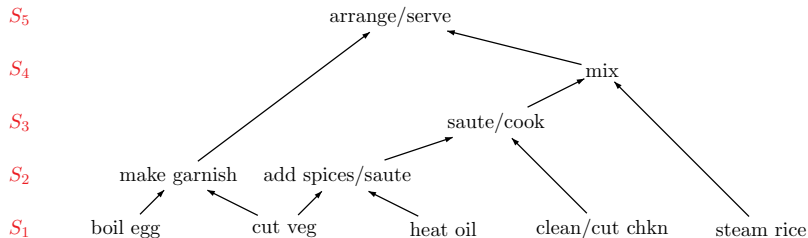
- ▶ Observe that we can schedule all of S_i at time i (since we know that all previous tasks were done earlier!).
- ▶ Thus, each S_i is an anti-chain.
- ▶ This solves the parallel task scheduling problem.

Parallel task scheduling and chains

Theorem

For a non-empty finite poset $(S, \preceq)$ with size of longest chain $= t$, we can partition S into t subsets $S_1, \dots, S_t$ such that $\forall i \in \{1, \dots, t\}, \forall a \in S_i$, if $b \preceq a, b \neq a$ then $b \in S_1 \cup \dots \cup S_{i-1}$.

Proof: Put each $a \in S$ in S_i such that i is the size of the largest chain ending at a .
Now proof by contradiction:



Parallel task scheduling and chains

Theorem

For a non-empty finite poset $(S, \preceq)$ with size of longest chain $= t$, we can partition S into t subsets $S_1, \dots, S_t$ such that $\forall i \in \{1, \dots, t\}, \forall a \in S_i$, if $b \preceq a, b \neq a$ then $b \in S_1 \cup \dots \cup S_{i-1}$.

Proof: Put each $a \in S$ in S_i such that i is the size of the largest chain ending at a .
Now proof by contradiction:

- ▶ Suppose $\exists i, a \in S_i, b \preceq a, b \neq a$ but $b \notin S_1 \cup \dots \cup S_{i-1}$.
- ▶ By defn of S_i , $\exists$ chain of size at least i ending at b .
- ▶ But now, $b \preceq a, b \neq a$ implies we can extend the chain to chain of size $\geq i + 1$, ending at a .
- ▶ But then a cannot be in S_i . Contradiction. □

Consequences for chains and anti-chains

Since each S_i was an anti-chain, a celebrated result follows...

Corollary (Mirsky's theorem, 1971)

If the largest chain in a poset $(S, \leq)$ is of size t , then S can be partitioned into t anti-chains.

Exercise: Apply this on the poset $(\mathcal{P}(S), \subseteq)$, where $S = \{1, 2, 3\}$. What does it imply for this poset?

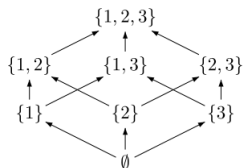
Another corollary (Dilworth's Lemma)

For all $t > 0$, any poset with n elements must have

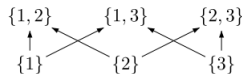
- ▶ either a chain of size greater than t
- ▶ or an antichain with at least $\frac{n}{t}$ elements.

Exercise: Prove it!

Minimal and maximal elements



Poset $P_1 = (S_1, \subseteq)$
where $S_1 = \mathcal{P}(\{1, 2, 3\})$



Poset $P_2 = (S_2, \subseteq)$
where $S_2 = \mathcal{P}(\{1, 2, 3\}) \setminus \{\emptyset, \{1, 2, 3\}\}$

Let $(S, \preceq)$ be a poset.

- ▶ $a \in S$ is **minimal** in S if $\forall b \in S, b \preceq a \implies b = a$
- ▶ $a \in S$ is **maximal** in S if $\forall b \in S, a \preceq b \implies a = b$.
- ▶ $a \in S$ is **least/minimum** element of S if $\forall b \in S, a \preceq b$
- ▶ $a \in S$ is **greatest/maximum** element of S if $\forall b \in S, b \preceq a$.

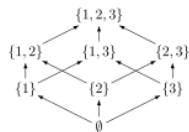
Examples: $\emptyset$ is a minimal and the minimum element in P_1 , $\{1\}, \{2\}, \{3\}$ are all minimal elements in P_2 , but P_2 does not have any minimum element.

Exercise: What are the maximal/maximum elements in P_1, P_2 ?

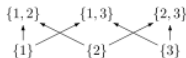
Least upper bounds and greatest lower bounds

Let $(S, \preceq)$ be a poset and $A \subseteq S$.

- ▶ $u \in S$ (resp. $l \in S$) is called an **upper bound** (resp. **lower bound**) of A iff $a \preceq u$ (resp. $l \preceq a$) for all $a \in A$.
- ▶ $u \in S$ is the **least upper bound (lub)** of A if it is an upper bound of A and is less than every other upper bound.
- ▶ $l \in S$ is the **greatest lower bound (glb)** of A if it is an lower bound of A and is greater than every other lower bound.



Poset $P_1 = (S_1, \subseteq)$
where $S_1 = \mathcal{P}(\{1, 2, 3\})$



Poset $P_2 = (S_2, \subseteq)$
where $S_2 = \mathcal{P}(\{1, 2, 3\}) \setminus \{\emptyset, \{1, 2, 3\}\}$

- ▶ Let $A = \{\{1\}, \{2\}\}$. Then $\{1, 2\}, \{1, 2, 3\}$ are upper bounds of A in P_1 and $\{1, 2\}$ is the lub of A . In P_2 , is there an lub for S_2 ?

Least upper bounds and greatest lower bounds

Let $(S, \preceq)$ be a poset and $A \subseteq S$.

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Some Observations (**Exercise:** Prove it!)

- ▶ The lub/glb of a subset A in S , if it exists, is unique.
- ▶ If the lub/glb of $A \subseteq S$ belongs to A , then it is the greatest/least element of A .

Lattices

Definition

- ▶ A **lattice** is a poset in which every pair of elements has both a lub and a glb (in the set), i.e., $\forall x, y \in S$, there exists $l, u \in S$ such that l is the glb and u is the lub of $\{x, y\}$.
- ▶ $(\mathcal{P}(S), \subseteq)$ is a lattice.
- ▶ What about $(\{2, 4, 5, 10, 12, 20, 25\}, |)$?

Applications of Lattices

- ▶ Models of information flow – think security clearance.
- ▶ Finite lattices have a strong link with Boolean Algebra
- ▶ Several other applications in many domains of mathematics and CS, including **formal semantics of programming languages, program verification.**